

Cross-Update Stress Tests for Cosmological-Constant Mechanisms

A Pre-Specified Framework Across Perturbative, EFT, and Physical-State Changes

1. THREE DISTINCT QUESTIONS

The cosmological-constant problem is often discussed as though it were one question. Proposed mechanisms, however, can be asked to answer at least three different ones.

The first is a protection or stability question: does a claimed relation survive the perturbative, effective-theory, or physical-state changes that the mechanism says it controls?

The second is an empirical-identification question: even if that protection is real, does the resulting physical prediction differ from the strongest admissible comparator strongly enough to identify the mechanism on the observables being tested?

The third is a numerical-realization question: does the theory actually claim to determine, restrict, or otherwise account for the realized numerical value of the cosmological term or another parameter? The existence of a measured value does not by itself establish a distinct physical mechanism whose job is to select that value. A value-selection mechanism is an additional scientific claim and must be specified and tested as such.

These questions do not rise and fall together. A protection result does not imply empirical identification. Empirical compatibility does not establish a value-selection claim. Failure of one proposed cosmological completion does not erase an independently established protection result.

This paper therefore asks a narrower cross-update question. For a specified mechanism M , does one protected physical relation survive the kinds of change that M itself claims to control, and where the mechanism predicts a response to a real physical successor state, is that response distinguishable from the strongest comparator?

That question is smaller than “does M solve the cosmological-constant problem?” The framework does not derive the observed value of Λ , identify dark energy with vacuum energy, assume that a compact-object response transfers to homogeneous cosmology, infer

physical truth from formal consistency, or infer a missing value-selecting mechanism merely because a realized parameter has a numerical value.

The three update tests used below are R_1 , for perturbative refinement; R_2 , for EFT threshold and matching changes; and S , for a real physical change of state. They can support one joint claim only when they concern the same protected physical job.

2. WHAT A MECHANISM MUST SPECIFY

A candidate mechanism first specifies a common protected functional,

$$\Pi [X; \theta_0],$$

where Π is the quantity, constraint, relation, or source-to-geometry response claimed to be protected; X is the physical input class relevant to that claim; and θ_0 is the theory data at a declared reference specification. Π need not be a scalar or a local operator. It may be an effective cosmological term, a covariant field-equation constraint, a global constraint, a filtering relation, or another physically evaluable response. What matters is that it performs the same physical job across every test included in a joint claim.

Every datum on which Π depends must also be specified. Integration constants, global variables, boundary or initial data, fitted offsets, and externally fixed constants are not interchangeable. If protection is obtained only by re-selecting a free datum after each update, the claim is fit-only. If the mechanism's own equations or constraints fix the relevant datum without post-update re-selection, the protection can be non-vacuous even when the resulting prediction is observationally equivalent, on a declared set of observables, to a reference theory with one fitted constant.

For claims involving vacuum-offset sensitivity, the mechanism must state its response in the matter-coupled frame to a uniform source shift,

$$T_{\mu\nu} \rightarrow T_{\mu\nu} + \lambda g_{\mu\nu}.$$

That response may be invariant, explicitly responsive, zero-point referenced, absorbed by a free datum, unresolved, or not applicable. A uniform shift is not sufficient for every mechanism. A physical successor may instead be localized, time-dependent, gradient-sensitive, symmetry-breaking, environmentally screened, or nonlocal. Every S test therefore specifies the actual physical variation ΔX used by that laboratory and shows from the mechanism's equations how the claimed protection answers it.

Legitimate running and environmental dependence must also be separated from retuning. A reference point θ_0 , or a region Θ_0 fixed in advance, is chosen before the tests are evaluated. Branch-specific values are then inherited through maps supplied by the theory,

$$\theta_i = F_i(\theta_0),$$

which may include renormalization-group running, EFT matching, state dressing, screening, global constraints, or boundary evolution. A joint result requires one common preimage of the successful regions under those maps. Independent parameter choices made separately for R_1 , R_2 , and S do not constitute one joint result.

The same requirement applies to the bridge between scales and states. The mechanism must specify what is carried, which equations or constraints carry it, which theory data are inherited, which degrees of freedom are integrated out or activated, and what uncertainty is introduced. Where successive maps are meaningful, composition should agree within the stated approximation:

$$B(s_3 \leftarrow s_1) \approx B(s_3 \leftarrow s_2) \circ B(s_2 \leftarrow s_1).$$

No universal UV-to-IR beta-function chain is required. A classical global constraint can be a legitimate bridge. The requirement is narrower: the same protection claim must be carried by the actual theory rather than replaced by a new independent coupling introduced for one convenient test.

3. THREE UPDATE TESTS

R_1 asks whether Π survives a controlled increase in perturbative completeness for the same physical target. The theory, background, renormalization prescription, calculation orders, allowed running, uncertainty treatment, and protection criterion are fixed before the higher-order result is evaluated. R_1 passes when the protected relation remains inside that criterion without post hoc retuning. It fails when the controlled correction violates the claim. If the higher-order calculation, target identity, or uncertainty is not controlled well enough to decide, the result remains unresolved.

R_2 asks the corresponding question across a change of effective resolution. A higher-resolution theory, heavy sector, matching prescription, low-energy target, inherited theory data, truncation uncertainty, and protected relation are fixed. R_2 passes when the matched low-energy account carries the same protection without introducing a new independent adjustment. It fails when valid matching destroys the relation. Uncontrolled matching or an ambiguous target leaves the question unresolved.

S differs in kind. It asks whether the same claim survives, or responds as predicted, through a real physical change of state. There is no universal S apparatus. The mechanism must identify the pre-state, successor state, physical transition, relevant source object, actual variation, and result rule before the target result is examined, together with every forward-model, comparator, alternative-coverage, sensitivity, data, or other burden made live by the registered claim and route. Requiredness must be fixed prospectively, including any conditional trigger. Sensitivity becomes a live burden only where exact absorption has been excluded and a nonzero residual survives the admitted comparator analysis; serious-alternative coverage is required only to the extent warranted by the identification claim being made. The physical change must alter an input relevant to Π ; the mechanism must apply in the proposed environment; a valid prediction must exist under M where the claim requires one; and exact absorption into the comparator must be tested before a deviation is called identifying.

The physical source object must also be well defined. A mechanism acting on total stress-energy, its trace, pressure, density, or another operational state quantity does not require a separate vacuum/matter split. A mechanism that acts on a special subcomponent must define that component before looking at the target data, show that it has physical identity beyond bookkeeping, preserve that identity under the relevant reparameterizations, and state what would falsify the identification. If the component is only model-conditional, the conclusion remains model-conditional. Global, averaged, nonlocal, or homogeneous-background mechanisms require laboratories that can access those structures; a local compact object is outside their domain unless the theory itself supplies a relevant local consequence.

Several distinct failure modes should not be collapsed. A proposed test can fail before data are taken because a required source object, shift response, bridge, or theory-data specification is missing. A well-defined test may still probe the wrong physical regime, or the mechanism may make no prediction for it. A valid prediction may be exactly degenerate with the strongest comparator. Only after those questions are cleared does inadequate sensitivity, forward-model uncertainty, data quality, or uncontrolled systematics become the limiting issue. Exact degeneracy is not the same as inadequate sensitivity, and degeneracy on one declared set of observables is not a claim of universal equivalence.

What counts as empirical success also depends on the claim. If M predicts an observable deviation outside the reference envelope, compatibility with M is not enough; the deviation must discriminate M from the reference and serious alternatives. If M predicts protection while an unprotected comparator predicts a distinguishable change, the experiment must be capable of seeing that change and must exclude it. “Nothing happened” does not establish

protection when the alternative could not have been seen. A parameter bound remains a bound unless the predeclared claim makes it more.

The kind of support matters as well. An exact identity or constraint, a controlled calculation, and a measurement are different forms of evidence. In particular, theoretical protection does not imply empirical discrimination. If R_1 and R_2 both pass because the same exact identity removes a constant contribution, they discharge two different update tests but do not count as two independent mechanisms of evidence.

4. CHOOSING A PHYSICAL-SUCCESSOR LABORATORY

The S laboratory follows from the mechanism rather than from the framework. Local trace- or pressure-dependent laws may make compact objects useful. A mechanism tied to an early-universe transition may require cosmological history. A global or historic constraint naturally routes to global cosmology. A homogeneous-only self-adjustment mechanism may make a neutron star irrelevant by construction.

Neutron stars are therefore conditional laboratories, not generic measurements of vacuum energy. They are useful only when the mechanism supplies explicit local dense-matter field equations and a forward prediction at neutron-star scales. The comparator should include reference gravity with the strongest justified microphysical and nuisance freedom, together with serious alternative nonstandard responses when they can mimic the same observables.

The central question is not whether the mechanism changes a mass-radius curve. It is whether an admissible reference model can reproduce the full prediction over the prespecified observables. If the answer is yes exactly, the mechanism is not identifiable on those observables. If a residual survives the reference fit but lies below forecast sensitivity, the limitation is one of reach rather than exact degeneracy. A new observable helps only when it probes physical structure not already absorbed by the comparator. Mass-radius, tidal, rotational, merger, cooling, and approximately EOS-insensitive relations therefore have to earn their role mechanism by mechanism; none is assumed to break gravity/EOS degeneracy automatically.

This scope discipline is consistent with existing neutron-star precedents. Gravitational-aether models have been proposed for testing in high-pressure neutron-star interiors, where the local response is pressure-sensitive.¹ Ventagli et al. have also modeled a QCD and vacuum-energy transition within general relativity and propagated it into mass-radius and

¹ F. Kamiab and N. Afshordi, “Neutron stars and the cosmological constant problem,” *Physical Review D* 84, 063011 (2011), DOI: 10.1103/PhysRevD.84.063011.

tidal observables.² These examples show that compact objects can test concrete mechanism-specific claims. They do not establish a model-independent dense-QCD vacuum component for use by every cosmological-constant mechanism. The broader dense-matter literature likewise supports treating high-density microphysics and its uncertainties as part of the comparator rather than as a solved input.³

5. CASE STUDY: GLOBAL VACUUM-ENERGY SEQUESTERING

As a worked case, we apply the framework to the original global Kaloper–Padilla vacuum-energy sequestering construction.⁴ The example is useful because it separates a genuine protection result from empirical mechanism identification without requiring a local compact-object interpretation.

In the original construction, global variables and a historic spacetime-volume constraint act on a protected matter sector coupled to a rescaled metric. At the declared matter-sector scope, constant vacuum-energy contributions are removed from the gravitational source by the same global projection, including matter-loop corrections under the construction’s stated assumptions.⁵ R_1 therefore passes at the level of the theory’s protection relation for the matter-loop refinement tested here. R_2 likewise passes for constant protected-matter threshold contributions carried by that same relation. Because both results arise from the same mechanism, they are not independent confirmations. These statements do not automatically extend to graviton loops; later work enlarges the sequestering construction to address that separate problem.⁶

The physical-successor result is different. After the protected constant vacuum contribution is removed, the original global field equation is

$$M_{\text{Pl}}^2 G^{\mu\nu} = \tau^{\mu\nu} - (1/4)\delta^{\mu\nu} \langle \tau^\alpha{}_\alpha \rangle,$$

with

² G. Ventagli, P. G. S. Fernandes, A. Maselli, A. Padilla, and T. P. Sotiriou, “Neutron stars and the cosmological constant problem,” *Physical Review D* 111, 024001 (2025), DOI: 10.1103/PhysRevD.111.024001.

³ R. Kumar et al., “Theoretical and experimental constraints for the equation of state of dense and hot matter,” *Living Reviews in Relativity* 27, 3 (2024), DOI: 10.1007/s41114-024-00049-6.

⁴ N. Kaloper and A. Padilla, “Sequestering the Standard Model Vacuum Energy,” *Physical Review Letters* 112, 091304 (2014), DOI: 10.1103/PhysRevLett.112.091304; “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), DOI: 10.1103/PhysRevD.90.084023.

⁵ N. Kaloper and A. Padilla, “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), especially Sec. III, DOI: 10.1103/PhysRevD.90.084023.

⁶ N. Kaloper and A. Padilla, “Vacuum Energy Sequestering and Graviton Loops,” *Physical Review Letters* 118, 061303 (2017), DOI: 10.1103/PhysRevLett.118.061303.

$$\Lambda_{\text{eff}} = (1/4)\langle\tau^{\alpha}_{\alpha}\rangle.^7$$

For one fixed history allowed by the model, the local equation is therefore general relativity with the same local stress-energy history and one spacetime-constant term Λ_{eff} . A matched reference theory using the same local matter history, the same realized local initial and boundary data, and $\Lambda_{\text{fit}} = \Lambda_{\text{eff}}$ reproduces observables determined entirely by that shared local history. That exact equivalence on the matched channels is a comparison result derived here from the cited field equations.

This adjudicates the two Standard-Model historical-successor tests considered in this paper. QCD thermodynamics can produce real early-universe transfer structure, including the smooth spectral effect associated with the continuous QCD crossover,⁸ but that structure is shared by both matched descriptions. The QCD history therefore does not identify original global sequestering on this channel. This is exact degeneracy on the comparison, not a sensitivity limit.

The electroweak test ends the same way. With the measured Higgs mass, the Standard Model has no bulk electroweak phase transition; its thermodynamics instead exhibits a crossover with a soft point near 160 GeV.⁹ Those thermodynamic effects can enter precision cosmological transfer calculations.¹⁰ But replacing a toy transition with realistic electroweak thermodynamics changes the shared stress-energy history in both descriptions; it does not create an electroweak-local adjustment law unique to original global sequestering. The electroweak history therefore does not identify the mechanism on this channel either.

Local non-identifiability does not make the global construction fit-only. Its global constraints restrict which histories are admitted. The original papers connect nonzero protected-sector scales to finite spacetime volume, eventual recollapse, and a transient phase of acceleration with w_{DE} approximately -1.¹¹ Explicit studies of the cosmological dynamics likewise show that the sequestering solution space is not identical to the unrestricted GR solution space.¹²

⁷ N. Kaloper and A. Padilla, “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), Sec. III, Eqs. (18)–(19), and discussion of whole-history determination, DOI: 10.1103/PhysRevD.90.084023.

⁸ K. Saikawa and S. Shirai, “Primordial gravitational waves, precisely: The role of thermodynamics in the Standard Model,” *Journal of Cosmology and Astroparticle Physics* 05 (2018) 035, DOI: 10.1088/1475-7516/2018/05/035.

⁹ M. Laine and M. Meyer, “Standard Model thermodynamics across the electroweak crossover,” *Journal of Cosmology and Astroparticle Physics* 07 (2015) 035, DOI: 10.1088/1475-7516/2015/07/035.

¹⁰ K. Saikawa and S. Shirai, “Primordial gravitational waves, precisely: The role of thermodynamics in the Standard Model,” *Journal of Cosmology and Astroparticle Physics* 05 (2018) 035, DOI: 10.1088/1475-7516/2018/05/035.

¹¹ N. Kaloper and A. Padilla, “Sequestering the Standard Model Vacuum Energy,” *Physical Review Letters* 112, 091304 (2014), DOI: 10.1103/PhysRevLett.112.091304; “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), DOI: 10.1103/PhysRevD.90.084023.

¹² P. P. Avelino, “Vacuum energy sequestering and cosmic dynamics,” *Physical Review D* 90, 103523 (2014), DOI: 10.1103/PhysRevD.90.103523.

The distinction is between local equivalence after a history is fixed and global restriction of the history that is admitted.

Those global restrictions support genuine one-way tests. In the large, old, approximately FRW branch, the 2014 framework paper shows that once cutoff-scale boundary terms are subdominant, the condition $\langle R \rangle = 0$ requires $k = +1$.¹³ A cosmological determination incompatible with positive spatial curvature would therefore falsify that scoped branch. Positive spatial curvature would be compatible with the branch, but would not identify the mechanism because ordinary GR also admits positively curved histories.

The most direct mechanism-facing test would evaluate the historic relation itself:

$$\Lambda_{\text{eff}} = (1/4) [\int_M d^4x \sqrt{-g} \tau^\alpha{}_\alpha] / [\int_M d^4x \sqrt{-g}].^{14}$$

But M is the complete spacetime history. The original construction therefore requires whole-history information, including future history, to evaluate this relation directly.¹⁵

That test is not presently measurable. The limitation is physical access to the integration domain, not merely instrumental precision.

A separate completion question asks whether the original finite-history construction can be supplied with viable physics that produces the required future collapse while fitting the observed cosmology. Kaloper and Padilla proposed a protected-sector scalar with a linear potential for this job.¹⁶ Ferreira and Avelino later found no observationally compatible solution when that same field is required to drive both the present accelerated epoch and the eventual collapse.¹⁷ That is a failure of the proposed completion at its stated joint role, not a failure of the already-separate matter-sector protection result.

The completion analysis also sharpens the identification boundary. If a completion adds ordinary local fields while preserving the original local metric equation with one historic constant, a realized history allowed by the model remains reproducible by GR with the same local fields, the same realized local data, and a matched cosmological term. The global

¹³ N. Kaloper and A. Padilla, “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), Sec. IV.C, especially Eq. (47) and the discussion immediately following it, DOI: 10.1103/PhysRevD.90.084023.

¹⁴ N. Kaloper and A. Padilla, “Vacuum energy sequestering: The framework and its cosmological consequences,” *Physical Review D* 90, 084023 (2014), Sec. III, Eqs. (18)–(19), and discussion of whole-history determination, DOI: 10.1103/PhysRevD.90.084023.

¹⁵ Ibid.

¹⁶ N. Kaloper and A. Padilla, “Sequestration of Vacuum Energy and the End of the Universe,” *Physical Review Letters* 114, 101302 (2015), DOI: 10.1103/PhysRevLett.114.101302.

¹⁷ R. Z. Ferreira and P. P. Avelino, “The fate of a Universe driven by a linear potential,” *Classical and Quantum Gravity* 35, 175018 (2018), DOI: 10.1088/1361-6382/aad474.

constraint may still remove freedom that GR leaves open, so a completion can make risky pre-data predictions and can be falsified. Successful observation of the predicted history, however, would not by itself identify the global constraint from the local observables.

Later sequestering constructions change material parts of the mechanism and should therefore be tested separately rather than imported as repairs of the original global model. The manifestly local formulation, for example, replaces the rigid global implementation with a local auxiliary-field formulation and leaves an a priori arbitrary finite radiatively stable contribution to background curvature;¹⁸ the graviton-loop construction enlarges the protected burden further.¹⁹

The case-study result is therefore bounded. Original global sequestering earns a matter-sector protection result under the assumptions tested here. Its QCD and electroweak local-history tests do not identify it against the strongest matched GR comparator used here. Its global constraints remain scientifically restrictive and falsifiable. One explicit completion designed to preserve the original mechanism fails its intended joint cosmological role. Direct testing of the historic relation requires complete-history information not presently available.

6. REPRODUCIBILITY AND CLAIM LIMITS

A reproducible execution should publish enough information for another group to reconstruct both the physical comparison and the interpretation of its outcome. At minimum, that includes the exact mechanism and protected relation, the physical inputs, parameters and initial or boundary data, the update or state change being tested, the maps used across scales or states, the comparator family, the uncertainty treatment, the observables, the decision criteria, and the analysis code or calculation notebooks where applicable.

Formal checking can help verify that a released protocol applies its own definitions and conditional logic consistently. It cannot decide whether a physical source exists, an EFT calculation is valid, a transition occurred, a comparator is scientifically adequate, or a dataset is trustworthy. Logical consistency is not empirical evidence.

The case study adds two further distinctions. First, a prediction fixed by a global constraint can be scientifically non-vacuous even when a matched reference theory can reproduce the realized local history after its free data are chosen. Second, the fact that a parameter has a

¹⁸ N. Kaloper, A. Padilla, D. Stefanyszyn, and G. Zahariade, “A Manifestly Local Theory of Vacuum Energy Sequestering,” *Physical Review Letters* 116, 051302 (2016), DOI: 10.1103/PhysRevLett.116.051302.

¹⁹ N. Kaloper and A. Padilla, “Vacuum Energy Sequestering and Graviton Loops,” *Physical Review Letters* 118, 061303 (2017), DOI: 10.1103/PhysRevLett.118.061303.

realized numerical value does not, by itself, establish a distinct mechanism whose job is to select that value. If a theory makes such a claim, that claim must be specified and tested independently.

The claim limits therefore remain narrow. Passage of R_1 or R_2 establishes only survival of the stated protection challenge. A successful S test establishes only the physical-successor result on the observables and comparator actually tested. A global-consistency or completion test can restrict or falsify histories without thereby identifying the mechanism positively on local observables. None of those results, separately or jointly, derives the observed value of Λ , identifies dark energy with vacuum energy, establishes uniqueness of the mechanism, or proves technical naturalness under arbitrary UV completion.

For original global sequestering, the endpoint is asymmetric. The matter-sector protection claim survives the R_1/R_2 challenges used here. The QCD and electroweak local-history tests are not identifying. Global cosmological restrictions remain testable and completion-specific predictions can fail. The direct historic-average relation is the cleanest parent-mechanism test identified here, but it depends on the complete spacetime history and is not presently measurable.

The strongest statement supported by this paper is therefore not that one mechanism has solved the cosmological-constant problem. It is that one protected relation can be followed across distinct kinds of change while protection, empirical identification, predictive restriction, completion viability, and numerical realization remain separate scientific questions.

CONCLUSION

Cosmological-constant mechanisms often package different scientific jobs under one label such as protection, cancellation, screening, naturalness, or solution. The framework developed here separates those jobs before testing them. It asks what quantity or relation is protected, what data remain free, how the same physical job is carried across perturbative, EFT, and physical-state changes, whether the resulting prediction survives the strongest comparator, and whether any claim about the realized numerical value has actually been supplied by the theory rather than inferred from the existence of the value itself.

The result is not a new cosmological-constant mechanism or an empirical resolution of the cosmological-constant problem. The distinctions instead block four false inferences: protection need not imply empirical identification; global restriction need not produce a unique local signature; failure of one completion need not erase an independently established

protection result; and a measured value does not by itself establish a separate mechanism that selects it.

Original global sequestering makes those distinctions concrete. At the protected-matter scope, the R_1 and R_2 challenges considered here pass. The QCD and electroweak historical-successor tests do not identify the mechanism against matched GR because the same realized local history can be reproduced with the same stress-energy history and one matched cosmological term. The global construction nevertheless remains more restrictive than fit-only GR through its whole-history constraints and associated falsification and completion burdens. The published linear-potential completion fails its joint present-acceleration and future-collapse role. The direct historic-average relation remains the cleanest parent-mechanism discriminator identified here, but its evaluation requires the complete spacetime history and is not presently measurable.

The scientific stopping point is therefore explicit. The paper locates what has been protected, what can be falsified, what is not identifying on the tested observables, what completion has failed, and what physical information would be needed to test the remaining global relation directly. Other cosmological-constant mechanisms can be placed under the same cross-update discipline, but each must begin from its own protected object, comparator, scale/state bridge, and physical-successor predictions.

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Signed,

Rick